
Does spatial context transfer? Neighbourhood-aware cell typing in seminiferous tubules under tissue-organization and species shift

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Methods for annotating spatial transcriptomics data increasingly use a spot’s tissue
2 neighbourhood as input, but they are benchmarked within one tissue and con-
3 dition, so it is unknown whether the spatial signal they learn holds when tissue
4 organization changes. We build a small benchmark from the public Slide-seqV2
5 testis pucks of Chen et al. (2021): three wild-type mouse, three diabetic *ob/ob*
6 mouse, and two human pucks (252,898 beads), with cell-type labels from NMFreg
7 (non-negative matrix factorization regression). Using germ-cell stage annotation in
8 seminiferous tubules as the task, we compare an expression-only classifier with
9 neighbourhood-augmented features, a GraphSAGE model, and test-time smoothing
10 of the expression-only posterior, under held-out-puck, wild-type-to-*ob/ob*, and
11 mouse-to-human transfer. Spatial context adds about one macro-F1 point within
12 a species and up to three on human pucks. The *ob/ob* pucks are not a distribu-
13 tion shift for this task. Across species, expression-only accuracy drops by ten
14 points; neighbourhood-augmented features and test-time smoothing keep their
15 gains, whereas GraphSAGE, which matches them within a species, becomes the
16 worst model, losing five points and fifteen F1 points on round spermatids. The
17 gain from context tracks how informative the neighbourhood is relative to the bead
18 itself rather than the target tissue’s neighbourhood homogeneity. Fixed spatial
19 aggregation transfers across the shifts we tested where learned spatial features do
20 not.

21 1 Introduction

22 Most recent methods for annotating cells or beads in spatial transcriptomics use the tissue neigh-
23 bourhood of a spot as an additional input, either by concatenating a spot’s own expression with an
24 aggregate of its neighbours’ [Singhal et al., 2024, Varrone et al., 2024] or by learning a graph neural
25 network (GNN) over a spatial adjacency graph [Hu et al., 2021, Dong and Zhang, 2022, Long et al.,
26 2023]. Benchmarks of these methods [Yuan et al., 2024, Li et al., 2022] evaluate within one tissue
27 and one condition, so they do not tell us whether the spatial signal a model learns still holds when
28 tissue organization changes, which is the question that matters when a model trained on a reference is
29 used to annotate samples from other conditions, individuals, or species.

30 The testis is a natural place to ask it. Spermatogenesis proceeds inside seminiferous tubules along
31 a radial axis, from spermatogonia at the basement membrane through spermatocytes and round
32 spermatids to elongating spermatids at the lumen, and tubule cross-sections cycle through a fixed
33 sequence of stages [Hess and de Franca, 2008, Chakravorty et al., 2026]. The same layering exists in
34 mouse and human, but the human cycle differs in stage number and in how stages tile a cross-section.
35 Chen et al. [2021] produced Slide-seqV2 [Stickels et al., 2021] pucks of wild-type mouse testis, of

testis from leptin-deficient diabetic *ob/ob* mice, in which they report disrupted spatial organization of the tubules, and of normal human testis, each with bead-level cell-type labels from NMFreg [Rodrigues et al., 2019]. This gives three levels of distribution shift on one tissue with one platform and one labelling procedure: a held-out puck of the same condition, a condition that alters tissue organization, and a different species.

We ask how much spatial context improves bead-level annotation of germ-cell stage, and whether the improvement survives each shift, comparing an expression-only classifier against neighbourhood-augmented features, a graph neural network, and test-time smoothing of the expression-only posterior over the target puck’s spatial graph. The last is trained without any spatial input and uses the target tissue’s geometry only at inference, separating the value of a spatial prior from the transferability of learned spatial features.

2 Data and task

Pucks. We use the eight processed Slide-seqV2 pucks released by Chen et al. [2021]: three wild-type (WT) mouse, three *ob/ob* diabetic (DB) mouse, and two normal human (HU) pucks, each with a bead-by-gene count matrix, bead coordinates, and a per-bead label over nine types (spermatogonia SPG, spermatocytes SPC, round spermatids RS, elongating/elongated spermatids ES, Sertoli, myoid, Leydig, endothelial, macrophage) assigned by NMFreg, a non-negative matrix factorization regression against a single-cell reference. We keep beads with a label and at least 100 unique molecular identifiers (UMIs), leaving 25,377 to 39,181 beads per puck (252,898 in total; Table 2).

Cross-species gene space. All pucks are restricted to one-to-one mouse–human orthologs from Mouse Genome Informatics (MGI) [Baldarelli et al., 2024] detected in every puck (13,573 genes), so a model trained on mouse can be applied to human without adaptation. Counts are library-size normalized and log-transformed [Wolf et al., 2018]. We select 2,000 highly variable genes on the WT pucks only [Stuart et al., 2019], then standardize each gene within each puck using that puck’s own mean and standard deviation. This unsupervised per-puck standardization is the only batch handling we apply.

Task. The primary task is four-class germ-cell stage annotation (SPG, SPC, RS, ES) on germ-cell beads, which are 81 to 90% of beads. This is the layer the radial tubule axis defines, and the reproducible part of the NMFreg annotation: within-puck and cross-puck macro-F1 agree (0.71 vs. 0.70), whereas the rare somatic types (endothelial, macrophage, myoid) are near chance from a single bead’s expression under any split. Six-class and nine-class variants are in the appendix.

Shift settings. Training pucks are always disjoint from the test puck. *WT*→*WT*: train on two WT pucks, test on the third (three folds). *WT*→*ob/ob*: train on all three WT pucks, test on each DB puck. *WT*→*human*: train on all three WT pucks, test on each human puck; *mouse*→*human* adds the DB pucks to training. As in-distribution references for the shifted targets, *ob/ob*→*ob/ob* and *human*→*human* train on the other pucks of the same condition. The NMFreg labels were computed from expression alone, so a gain from spatial features cannot be an artefact of how the labels were made. For each puck we also compute neighbourhood homogeneity, the mean fraction of a bead’s 15 nearest spatial neighbours that share its label; it is used only to describe the pucks, never as an input. Among germ-cell beads it is 0.53 to 0.56 on WT, 0.49 to 0.57 on *ob/ob*, and 0.38 to 0.40 on human pucks.

3 Models and evaluation protocol

All models operate on a 50-dimensional principal component analysis (PCA) embedding z_i of the standardized expression, with the PCA fit on the training pucks only and applied to the test puck. Let $\mathcal{N}_k(i)$ be the k nearest spatial neighbours of bead i within its puck, and $\bar{z}_i^{(k)} = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} z_j$ the neighbourhood mean, computed over all beads of the puck (non-germ included), since context is label-free.

Table 1: Macro-F1 for germ-cell stage under six settings (left three in-distribution, right three shifted); mean $\pm$ standard deviation (SD) across test pucks. nbr: mean embedding of the 15 nearest spatial neighbours. α selected on the WT $\rightarrow$ WT folds. MLP and neighbourhood-only rows: Table 3.

Model	WT $\rightarrow$ WT	ob/ob $\rightarrow$ ob/ob	human $\rightarrow$ human	WT $\rightarrow$ ob/ob	WT $\rightarrow$ human	mouse $\rightarrow$ human
LR (expr)	0.699 $\pm$ 0.013	0.717 $\pm$ 0.022	0.486 $\pm$ 0.008	0.713 $\pm$ 0.026	0.385 $\pm$ 0.005	0.385 $\pm$ 0.006
LR (expr+nbr)	0.711 $\pm$ 0.013	0.723 $\pm$ 0.016	0.513 $\pm$ 0.011	0.721 $\pm$ 0.023	0.398 $\pm$ 0.007	0.396 $\pm$ 0.008
LR (expr) + smooth ($\alpha=0.5$)	0.712 $\pm$ 0.016	0.733 $\pm$ 0.019	0.501 $\pm$ 0.008	0.732 $\pm$ 0.021	0.401 $\pm$ 0.005	0.400 $\pm$ 0.006
GraphSAGE	0.711 $\pm$ 0.015	0.716 $\pm$ 0.023	0.505 $\pm$ 0.013	0.721 $\pm$ 0.023	0.347 $\pm$ 0.012	0.350 $\pm$ 0.012

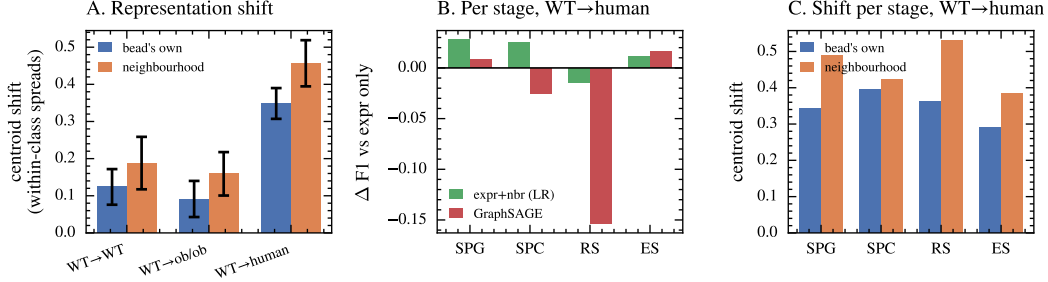


Figure 1: (A) Distance each germ stage’s centroid moves from training to test puck, in units of its within-class spread in training, for the bead’s own 50-dimensional embedding and for the neighbourhood mean; bars are means over stages and test pucks, error bars their standard deviation. (B) Per-stage F1 change relative to the expression-only LR, and (C) per-stage centroid shift, both under species shift. Round spermatids are where GraphSAGE loses most and where the neighbourhood representation moves furthest.

84 **Expression only.** A multinomial logistic regression (LR) on z_i , and a two-layer multilayer percep-
85 tron (MLP; 256 hidden units, dropout 0.2). Both use class-balanced loss weights.

86 **Neighbourhood-augmented features.** LR and MLP on $[z_i, \bar{z}_i^{(k)}]$, the construction used by
87 BANKSY [Singhal et al., 2024] and related methods, for $k \in \{5, 15, 30\}$. As a diagnostic we
88 also fit LR on $\bar{z}_i^{(k)}$ alone, i.e. on the surroundings without the bead.

89 **Graph neural network.** A two-layer GraphSAGE with mean aggregation [Hamilton et al., 2017],
90 $h'_i = \sigma(W_1 h_i + W_2 \bar{h}_i^{(15)})$, 128 hidden units, trained full-batch on the union of training pucks (each
91 puck is its own connected graph) and applied to the test puck’s graph.

92 **Test-time smoothing.** We take the expression-only LR posterior p_i on the test puck and replace it
93 with $p'_i = (1 - \alpha) p_i + \alpha \bar{p}_i^{(15)}$, then predict $\arg \max p'_i$. This is one step of label propagation on the
94 model’s own outputs [Huang et al., 2021]. We sweep $\alpha \in \{0.25, 0.5, 0.75, 1\}$.

95 **Evaluation.** Macro-F1 over the classes present in the test puck and per-class F1, averaged over
96 seeds (three for MLP and GraphSAGE; LR is deterministic) and then over the test pucks of a setting,
97 with the standard deviation across test pucks. Hyperparameters were fixed a priori and shared across
98 all settings, and nothing was tuned on any test puck; the values are listed in the appendix. The full
99 grid runs on two CPU cores in a few hours.

100 4 Results

101 **In-distribution gains are small and consistent.** On held-out WT pucks the expression-only logistic
102 regression reaches 0.699 macro-F1, and every form of spatial context adds about one point: 0.711 for
103 neighbourhood-augmented LR and for GraphSAGE, 0.712 for test-time smoothing (Table 1). The
104 gain is small but systematic, positive on all 15 setting–test-puck combinations. A neighbourhood-only
105 classifier reaches 0.636, so most of the stage signal is in the bead itself. Gains are similar on ob/ob and
106 larger on human folds, where the expression-only ceiling is lower (0.486) and neighbourhood features

add 2.7 points (0.513). Neighbourhood size matters little (at most 0.008 across $k \in \{5, 15, 30\}$), and an MLP on the same features is 0.6 to 2.1 points below the LR in every setting (Tables 3, 4).

The *ob/ob* shift is small for this task. Models trained on WT pucks perform as well on *ob/ob* pucks as models trained on other *ob/ob* pucks (0.713 vs. 0.717 expression only; 0.721 vs. 0.723 with neighbourhood features). The spatial gains carry over unchanged, and germ-cell homogeneity overlaps between conditions (0.49 to 0.57 vs. 0.53 to 0.56). A classifier trained to tell WT from *ob/ob* beads inside the models’ feature space performs at chance (area under the curve, AUC: 0.499 own, 0.519 neighbourhood; Table 8), so the condition is not a covariate shift in the space the models see, and the disruption reported by Chen et al. [2021] does not register at this resolution.

Species shift is large and is dominated by expression. Expression-only macro-F1 falls from 0.486 (human→human) to 0.385 (WT→human), and adding the *ob/ob* pucks to training does not change it (0.385). Neighbourhood-augmented LR retains a gain under species shift (0.398, +1.3), as does test-time smoothing (0.401, +1.6). GraphSAGE does not: it is within 0.008 of neighbourhood-augmented LR in every in-distribution setting and becomes the worst model across species, at 0.347 (−3.8 vs. expression only, −5.1 vs. neighbourhood LR; seed-to-seed SD ≤ 0.007). Per stage (Fig. 1B), the loss is concentrated on round spermatids, the largest human class, whose F1 drops from 0.390 to 0.237 with GraphSAGE and to 0.376 with neighbourhood LR; with neighbourhood LR, spermatogonia and spermatocytes still gain about 3 points.

The neighbourhood representation is the less stable one. How far each stage’s centroid moves between training and test pucks, in units of its own within-class spread, separates the two feature spaces (Fig. 1A). Both are nearly stationary within a species (0.12 own, 0.19 neighbourhood) and under the *ob/ob* condition (0.09, 0.16), and both move across species (0.35, 0.46). In all 24 stage-by-puck comparisons the neighbourhood centroid moves further than the bead’s own, by factors of 1.06 to 3.30: a fixed average inherits that drift once, whereas a model learning a function of the neighbourhood distribution is fitted to it. Round spermatids show both the largest cross-species neighbourhood displacement (0.53) and the largest GraphSAGE loss (Fig. 1B,C), which with four stages we read as consistent with the mechanism rather than evidence for it.

Secondary observations. The gain from context does not track the target puck’s neighbourhood homogeneity: the largest gains occur on the human pucks, which have the lowest (0.38 to 0.40). It tracks instead how informative the neighbourhood is relative to the bead, a within-sample association rather than a usable predictor, since both come from the same labels (Fig. 3C). Test-time smoothing peaks at $\alpha = 0.5$ in every setting (Fig. 3D). At the bead level the within-species gain sits in the lowest-UMI tertile, and context lowers accuracy for beads with heterogeneous neighbourhoods (Table 7).

5 Discussion

The gains are small because the neighbourhood is largely redundant with the bead: a neighbourhood-only classifier recovers most of the signal the bead already carries. We would not expect this on niche-level labels, where the neighbourhood defines the class rather than duplicating it. Organization changes that matter at the scale of tubule architecture can be invisible at the scale of stage annotation, which is what the *ob/ob* comparison shows.

The more useful result is the split between how context is used. Fixed aggregation and post-hoc smoothing keep their gains under both shifts; a two-layer GNN matches them within a species and loses about five points across species. Both see the same neighbourhood mean, but only the GNN learns a function of it, and that is the less stable of the two representations. For reference-based annotation across species, a fixed spatial operator has an advantage that in-distribution benchmarks do not reveal.

Limitations. Labels are NMFreg assignments, not ground truth, and the human ones are noisier (in-distribution ceiling 0.49 vs. 0.70), so results measure agreement with a published annotation; the human estimates rest on two pucks. Stronger cross-species integration [Rosen et al., 2024, Lotfollahi et al., 2022] would likely narrow the species gap for all models; whether it would close the GNN’s gap is open. We evaluated one GNN architecture at one depth. Code and splits will be released.

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Table 2: Puck statistics after filtering. Homogeneity is the mean fraction of a bead’s k nearest spatial neighbours that share its NMFreg label, over all beads ($k=5, 15, 30$) and over germ-cell beads only ($k=15$).

Puck	Condition	Beads	Median UMI	Homog. $k=5$	$k=15$	$k=30$	Germ, $k=15$
WT1	WT mouse	29,178	719	0.524	0.474	0.432	0.538
WT2	WT mouse	29,607	782	0.524	0.470	0.426	0.533
WT3	WT mouse	39,181	604	0.561	0.507	0.464	0.561
DB1	<i>ob/ob</i> mouse	25,377	797	0.484	0.426	0.385	0.485
DB2	<i>ob/ob</i> mouse	35,629	426	0.537	0.477	0.433	0.530
DB3	<i>ob/ob</i> mouse	30,073	684	0.574	0.518	0.473	0.568
HU1	human	30,464	602	0.362	0.336	0.316	0.380
HU2	human	33,389	600	0.379	0.352	0.331	0.397

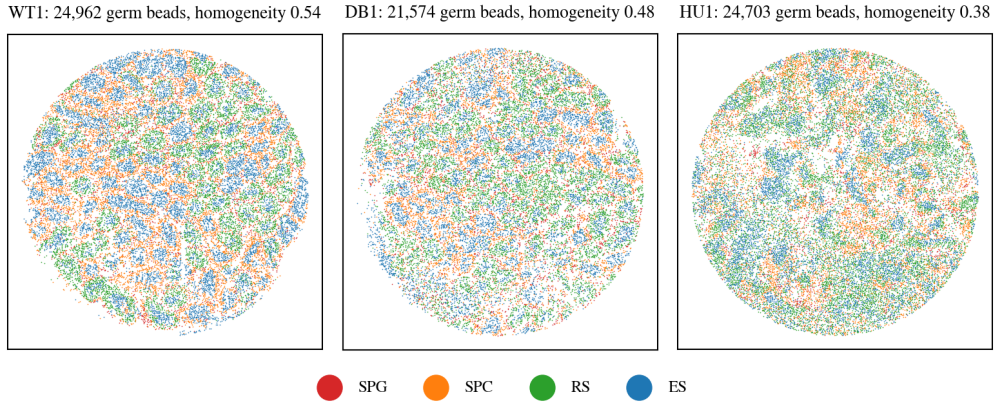


Figure 2: Germ-cell beads of one WT mouse, one *ob/ob* mouse, and one human puck, coloured by NMFreg stage. Tubule cross-sections with a spermatogonia/spermatocyte rim and a spermatid core are visible in mouse; human tubules are larger and stage labels are more mixed.

Table 3: Full version of Table 1, including the MLP models. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	<i>ob/ob</i> → <i>ob/ob</i>	human→human	WT→ <i>ob/ob</i>	WT→human	mouse→human
LR (expr)	0.699 $\pm$ 0.013	0.717 $\pm$ 0.022	0.486 $\pm$ 0.008	0.713 $\pm$ 0.026	0.385 $\pm$ 0.005	0.385 $\pm$ 0.006
LR (expr+nbr)	0.711 $\pm$ 0.013	0.723 $\pm$ 0.016	0.513 $\pm$ 0.011	0.721 $\pm$ 0.023	0.398 $\pm$ 0.007	0.396 $\pm$ 0.008
LR (nbr only)	0.636 $\pm$ 0.003	0.607 $\pm$ 0.019	0.464 $\pm$ 0.006	0.609 $\pm$ 0.029	0.365 $\pm$ 0.012	0.355 $\pm$ 0.008
LR (expr) + smooth ($\alpha=0.5$)	0.712 $\pm$ 0.016	0.733 $\pm$ 0.019	0.501 $\pm$ 0.008	0.732 $\pm$ 0.021	0.401 $\pm$ 0.005	0.400 $\pm$ 0.006
MLP (expr)	0.693 $\pm$ 0.014	0.704 $\pm$ 0.036	0.475 $\pm$ 0.012	0.704 $\pm$ 0.028	0.370 $\pm$ 0.009	0.369 $\pm$ 0.009
MLP (expr+nbr)	0.705 $\pm$ 0.018	0.708 $\pm$ 0.029	0.498 $\pm$ 0.014	0.713 $\pm$ 0.023	0.378 $\pm$ 0.013	0.375 $\pm$ 0.011
GraphSAGE	0.711 $\pm$ 0.015	0.716 $\pm$ 0.023	0.505 $\pm$ 0.013	0.721 $\pm$ 0.023	0.347 $\pm$ 0.012	0.350 $\pm$ 0.012

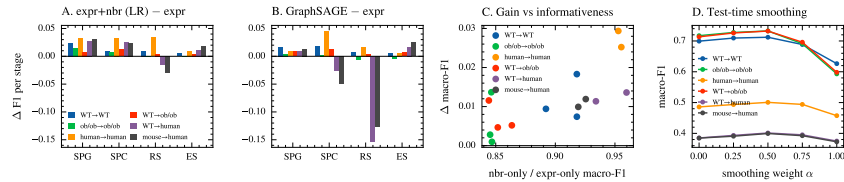


Figure 3: (A, B) Per-stage change in F1 from adding spatial context, relative to the expression-only LR, in all six settings; shared vertical scale. (C) Gain from neighbourhood features per test puck against the relative informativeness of context. (D) Test-time smoothing of the expression-only posterior against the weight α ; $\alpha = 1$ replaces each bead’s posterior by its neighbours’ mean.

Neighbourhood size and smoothing weight. Table 4 reports the LR with neighbourhood features for $k \in \{5, 15, 30\}$ and test-time smoothing for all α .

Table 4: Sensitivity to neighbourhood size k (LR on expression plus neighbourhood mean) and to the smoothing weight α (expression-only LR posterior smoothed over the test puck’s 15-nearest-neighbour graph). Macro-F1 for germ-cell stage, mean over test pucks.

Setting	expr+nbr, k			smoothing weight α				
	5	15	30	0	0.25	0.5	0.75	1
WT→WT	0.715	0.711	0.708	0.699	0.709	0.712	0.689	0.626
ob/ob→ob/ob	0.728	0.723	0.720	0.717	0.728	0.733	0.691	0.593
human→human	0.511	0.513	0.510	0.486	0.494	0.501	0.494	0.457
WT→ob/ob	0.722	0.721	0.718	0.713	0.726	0.732	0.696	0.598
WT→human	0.399	0.398	0.394	0.385	0.393	0.401	0.395	0.375
mouse→human	0.397	0.396	0.390	0.385	0.391	0.400	0.392	0.373

Six-class and nine-class labels. Tables 5 and 6 repeat the linear models on all beads with a six-class scheme (four germ stages, tubular somatic, interstitial) and with the original nine NMFreg labels. Macro-F1 is much lower for nine classes because the rare somatic types (endothelial, macrophage, myoid) are near chance from a single bead’s expression even in-distribution. The gains from context are smaller than for germ-cell stage (0.4 to 1.0 points within a species for six classes, none for nine classes) but again largest on human in-distribution folds (about two points); across species with nine classes, neither neighbourhood features nor smoothing changes macro-F1 by more than 0.004.

Table 5: Six-class annotation (SPG, SPC, RS, ES, tubular somatic, interstitial), all beads. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.524 $\pm$ 0.018	0.549 $\pm$ 0.010	0.418 $\pm$ 0.001	0.547 $\pm$ 0.009	0.301 $\pm$ 0.004	0.299 $\pm$ 0.006
LR (expr+nbr)	0.533 $\pm$ 0.017	0.553 $\pm$ 0.007	0.441 $\pm$ 0.002	0.553 $\pm$ 0.007	0.308 $\pm$ 0.003	0.306 $\pm$ 0.002
LR (nbr only)	0.453 $\pm$ 0.003	0.429 $\pm$ 0.013	0.391 $\pm$ 0.006	0.435 $\pm$ 0.017	0.280 $\pm$ 0.007	0.278 $\pm$ 0.007
LR (expr) + smooth ($\alpha=0.5$)	0.531 $\pm$ 0.018	0.558 $\pm$ 0.016	0.433 $\pm$ 0.000	0.561 $\pm$ 0.012	0.316 $\pm$ 0.006	0.315 $\pm$ 0.002

Table 6: Nine-class annotation with the original NMFreg labels, all beads. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.390 $\pm$ 0.005	0.405 $\pm$ 0.000	0.309 $\pm$ 0.002	0.409 $\pm$ 0.010	0.181 $\pm$ 0.006	0.182 $\pm$ 0.003
LR (expr+nbr)	0.392 $\pm$ 0.004	0.401 $\pm$ 0.004	0.330 $\pm$ 0.001	0.413 $\pm$ 0.012	0.177 $\pm$ 0.005	0.180 $\pm$ 0.004
LR (nbr only)	0.316 $\pm$ 0.004	0.296 $\pm$ 0.010	0.280 $\pm$ 0.005	0.305 $\pm$ 0.007	0.160 $\pm$ 0.006	0.168 $\pm$ 0.002
LR (expr) + smooth ($\alpha=0.5$)	0.393 $\pm$ 0.006	0.412 $\pm$ 0.010	0.328 $\pm$ 0.001	0.416 $\pm$ 0.014	0.179 $\pm$ 0.005	0.181 $\pm$ 0.003

Gain by sequencing depth and local homogeneity. Table 7 stratifies the accuracy gain from neighbourhood features by the test bead’s UMI tertile and by the fraction of its 15 neighbours that share its label. Within a species the gain comes from low-UMI beads; under species shift it comes from high-UMI beads. In every setting context lowers accuracy for beads with heterogeneous neighbourhoods (fewer than 40% same-label neighbours) and raises it for the rest, so the puck-level gain is a net of these two effects.

Table 7: Accuracy gain from neighbourhood features (expr+nbr minus expr, LR) stratified by the test bead’s UMI tertile within its puck and by its local label homogeneity. Mean over test pucks of each setting.

Setting	UMI tertile			local homogeneity				
	low	mid	high	0-0.2	0.2-0.4	0.4-0.6	0.6-0.8	0.8-1
WT→WT	+0.023	+0.003	+0.003	-0.023	-0.010	+0.022	+0.030	+0.018
ob/ob→ob/ob	+0.015	-0.001	-0.002	-0.036	-0.011	+0.017	+0.024	+0.019
human→human	+0.047	+0.021	+0.018	-0.048	+0.014	+0.091	+0.111	+0.089
WT→ob/ob	+0.015	+0.002	+0.003	-0.035	-0.009	+0.019	+0.030	+0.021
WT→human	-0.011	+0.002	+0.029	+0.000	-0.007	+0.011	+0.034	+0.050
mouse→human	-0.001	+0.004	+0.016	-0.007	-0.001	+0.013	+0.033	+0.042

Representation shift. Table 8 reports both measures of how far the feature distribution moves between training and test pucks: the area under the receiver operating characteristic curve (AUC) of a logistic regression trained to separate training-domain from test-domain beads, and the per-stage centroid displacement of Fig. 1A. Displacement is normalised by the within-class spread in that same space, since averaging over 15 neighbours shrinks the spread as well as the offset; the AUC is not normalised and is reported only to establish that the *ob/ob* condition sits at chance in both spaces.

Table 8: Representation shift between training and test pucks. Domain-classifier AUC (0.5 = indistinguishable) and mean per-stage centroid displacement in units of within-class spread, for the bead’s own embedding and for the neighbourhood mean.

Setting	Test	domain-classifier AUC		centroid shift	
		own	nbr	own	nbr
WT→WT	WT3	0.513	0.527	0.124	0.188
WT→ob/ob	DB1	0.499	0.534	0.067	0.148
WT→ob/ob	DB2	0.503	0.518	0.091	0.189
WT→ob/ob	DB3	0.495	0.504	0.116	0.141
WT→human	HU1	0.552	0.619	0.355	0.465
WT→human	HU2	0.551	0.620	0.342	0.448

Hyperparameters. All values were fixed before running the grid and shared across every setting; none were tuned on a test puck. Logistic regression: $C = 0.5$, class-balanced weights, at most 1000 iterations. MLP: 256 hidden units, dropout 0.2, AdamW with learning rate 10^{-3} and weight decay 10^{-4} , 40 epochs at batch size 512. GraphSAGE: two layers, 128 hidden units, dropout 0.2, AdamW with learning rate 3×10^{-3} and weight decay 10^{-4} , 150 full-batch epochs. PCA is fit on the training pucks only, with 50 components throughout.